

# EffectSlice: Evidence-Carrying Admission for Paper-Derived Procedural Cores

Anonymous submission

## Abstract

Turning a paper into a compact agent artifact does not make that artifact a reliable substitute: source provenance and behavioral sufficiency are different claims. We introduce **EffectSlice**, a proof-of-concept, reducer-agnostic candidate-admission protocol and evidence package for paper-derived procedural cores. Given a no-artifact baseline (B), registered full artifact (F), and selected strict subset (S), it binds instructions to paper spans and dependencies, freezes the candidate, permits visible public testing, and scores each fresh terminal run once with a private suite. Admission is a hash-bound finite-schedule decision, not a population estimate. Across two sequentially registered schedules and 234 fresh DeepSeek endpoint conversations, the protocol separates three decision states: SNAP prefix 03 has insufficient F and S (0/15/5 B/F/S successes, Reject); Toolformer prefix 01 has sufficient F but insufficient S (0/17/8, Reject); in a separately registered later schedule, a local admission is issued for pre-existing Toolformer prefix 04 (0/18/17, Admit) on a new nonoverlapping private registry. A planted redundancy smoke test and byte-identical identity instrument pass their registered internal checks. These task-local outcomes do not establish broad automatic reduction, paper-level equivalence, or human benefit.

## Introduction

Paper-reproduction agents increasingly turn research descriptions into code and executed experiments (Starace et al. 2025; Zhao et al. 2025), while paper-to-agent systems package papers and associated software as interactive tools (Miao et al. 2025). Agent-skill systems likewise compile external resources into reusable instructions (Shen et al. 2026; Pan et al. 2026). Both can pair vendor inference with local execution.

The difficult step is deciding what the reusable *core* is. A paper may describe a five-step mechanism, while a reducer retains only the first three steps because they were sufficient on one development example. The resulting artifact can be short and traceable to the paper without being a reliable substitute for the complete procedure. Conversely, a complete artifact can be correct but inconsistently used by a bounded

agent. These are behavioral questions that token counts and source overlap do not answer.

Recent work makes this tension concrete. SkillReducer reports that large-scale skill compression can improve functional quality (Gao et al. 2026). A controlled 1,200-rollout decomposition instead finds no representation strategy that improves over raw skills and identifies executor capability as the dominant factor (Xu and Wu 2026). Both are important optimization studies. Our question is downstream and candidate-specific: after any system proposes a paper-derived core, what evidence is required before a user is told that it can stand in for the complete paper-derived artifact?

Three requirements follow: source-addressed instructions and dependencies; public feedback separated from terminal private scoring; and registered positive, negative, and byte-identical controls. Provenance establishes identity, not a complete interpretation, while controls expose obvious gate or session failures.

We introduce **EffectSlice**, an evidence-carrying admission protocol rather than a new compression algorithm. It represents a paper-derived procedure as source-addressed atoms, freezes a dependency-closed strict subset, and runs three conditions: no artifact (B), the full artifact (F), and the selected slice (S). Each condition receives a fresh provider conversation. The public test is visible and callable; the private 64-case scorer is invoked only on the fixed terminal patch and its result is not returned for another model turn. Separate pre-run and post-run manifests bind the task, paper source, atoms, artifacts, tests, scorer, runner, transport, schedule, outputs, and analysis lineage by digest.

The evidence addresses three questions: whether realized B/F/S runs distinguish full-artifact insufficiency from subset-only insufficiency (RQ1); whether positive, negative, and identity controls exercise admission, rejection, and session-noise sanity states (RQ2); and whether a post-selected candidate can earn a valid local admission on a separately registered, nonoverlapping boundary (RQ3).

Our contributions are:

- We operationalize a source-addressed atom schema with exact source locations, executable roles, dependency edges, and immutable artifact identities, separating provenance from behavioral sufficiency.
- Our main contribution is the candidate-level admission record: candidate and source identities, public/private

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information boundary, frozen schedule, immutable outcomes, and a binary finite-schedule Admit/Reject decision. Reject means only “do not admit under this registered boundary,” and makes no IID, cross-paper, or population claim.

- Our proof of concept runs three non-planted prefix candidates plus positive, negative, and identity checks in two hash-registered schedules. V4 exposes full-artifact and subset-only insufficiency; fresh V5 evidence admits a pre-existing four-of-five-atom candidate. The package preserves all 234 conversations, patches, scores, and digests, and cross-checks two disclosed analyzer compatibility repairs.

We do not claim a general compression advantage, lower cost, local-model inference, full-paper reproduction, or benefit to human users.

## Related Work

**Paper reproduction and executable research.** PaperBench evaluates agents that recreate AI research repositories (Starace et al. 2025); AutoReproduce adds paper lineage to automated experiment reproduction (Zhao et al. 2025); and Paper2Agent builds interactive agents around papers and code, emphasizing reliable construction, iterative testing, and source linkage (Miao et al. 2025). We treat Paper2Agent as an upstream full-agent generator: EffectSlice starts after a registered full artifact exists and audits a later strict-subset substitution claim.

**Skill construction, retrieval, and compression.** SkillFoundry and Anything2Skill compile reusable skills from heterogeneous sources (Shen et al. 2026; Pan et al. 2026). Graph-of-Skills and SkillRAE organize dependency-aware retrieval and context compilation, while ClawTrace distills skill content from execution traces (Liu et al. 2026; Meng, Wang, and Fang 2026; Yuan et al. 2026). Related tool-learning work iteratively acquires tool use, preserves general capability during tool training, constructs reusable toolsets, and selects tools for agent execution (Gao et al. 2024; Hao et al. 2025; Ma et al. 2025; Jia and Li 2026). SkillReducer compresses routing descriptions and skill bodies, then restores content when task feedback shows a regression; it also reports an independent deterministic benchmark (Gao et al. 2026). Xu and Wu freeze ten skill-delivery conditions over 40 software tasks, compare raw, shortened, structured, and scoped forms, and use task-clustered inference and real-cost accounting (Xu and Wu 2026). Shaposhnikov et al. provide our strongest utility baseline by comparing behavior with and without whole skills (Shaposhnikov et al. 2026). Their Agentic Skills Framework measures whole-skill marginal utility; our contrast fixes a strict subset against a registered full-artifact adequacy control and binds the resulting release identity to source locators and the evaluation boundary. EffectSlice neither competes with their reducers nor repeats their cost question. It adds source-span binding and a candidate-level admission record after the candidate is frozen.

**Tool-agent evaluation.** MCP-AgentBench tests agents with MCP-mediated tools, while complementary work maps tool-use capability boundaries through cognitive load (Guo et al. 2026; Wang et al. 2026). EffectSlice narrows this evaluation question to whether one frozen strict subset may substitute for its registered full artifact, and retains source and release identities for that decision.

**Admission and risk control.** Selective prediction studies when a system should reject rather than make a prediction, typically reporting a risk-coverage trade-off (Geifman and El-Yaniv 2019). Conformal risk control calibrates a monotone loss against held-out data (Angelopoulos et al. 2022), and AgentBoard advocates fine-grained rather than final-success-only evaluation of multi-turn agents (Ma et al. 2024). EffectSlice does not claim a distribution-free risk guarantee: it records a finite, preregistered local admission decision with integrity checks, explicit controls, and its applicable task and release boundary.

The established components are reduction, holdout isolation, skill-utility measurement, provenance, and digest manifests. Our specific technical increment is their coupling into a candidate-specific substitution decision and release record. Despite its name, EffectSlice is an admission protocol, not a slicing algorithm.

**Reduction and adaptive evaluation.** Delta debugging and C-Reduce remove components while preserving observed behavior (Zeller and Hildebrandt 2002; Regehr et al. 2012); cause reduction extends the objective beyond failure-inducing inputs (Groce et al. 2016). Prompt compression trades context size against retained information (Jiang et al. 2023; Pan et al. 2024; Liskavets et al. 2025). Repeated holdout exposure can overfit an adaptively selected hypothesis (Dwork et al. 2015; Nakkiran and Błasiok 2018). EffectSlice applies this information-flow lesson to tool-using code agents: public tests support development, while the fixed patch receives one terminal private evaluation. The increment is binding that isolation to paper-source addresses, B/F/S candidate identities, and an archival local decision record.

## Problem Setting

Let paper  $P$  have a cryptographic identity  $d_P$ . We represent a bounded paper-specific procedure as atoms  $A = \{a_1, \dots, a_K\}$  and a directed acyclic dependency graph  $G$ . Each atom stores

$$a_i = (\text{id}_i, \ell_i, s_i, r_i, D_i), \quad (1)$$

where  $\ell_i$  is an exact source locator (document digest, byte and line spans),  $s_i$  is the executable instruction,  $r_i$  its contract role, and  $D_i$  the set of atom IDs of its direct dependencies. A candidate atom set  $A_S \subsetneq A$  is nonempty and dependency closed iff  $D_i \subseteq \{\text{id}_j : a_j \in A_S\}$  for every  $a_i \in A_S$ . A deterministic renderer  $R$ , using topological order with atom-ID tie breaking, produces the full artifact  $F = R(A)$  and candidate artifact  $S = R(A_S)$ . Both are registered before confirmation. This representation records provenance; it does not assert semantic completeness or that an agent will use the instructions.

| Study                                               | Artifact and objective                                                                 | Evaluation boundary                                                                                          | Distinguishing output                                                |
|-----------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| SkillReducer (Gao et al. 2026)                      | Generic agent skills; token-efficient routing and progressive disclosure               | Self-correcting task feedback plus an independent deterministic skill benchmark                              | Optimized skill and aggregate retention/quality                      |
| Xu and Wu (Xu and Wu 2026)                          | Raw and transformed skills; quality/cost factor decomposition                          | 40 tasks, 10 conditions, three repetitions; task-clustered contrasts                                         | Multi-task aggregate condition comparison and cost diagnosis         |
| Agentic Skills Framework (Shaposhnikov et al. 2026) | Whole executable skills; marginal utility                                              | Scalable with-skill versus without-skill task evaluation                                                     | Whole-skill utility across task/model configurations                 |
| EffectSlice (ours)                                  | Source-addressed atoms from each source paper; admit or reject one fixed strict subset | Visible public test, one terminal private score per fresh condition run, complete registered finite schedule | Hash-bound candidate record with a task-local admission or rejection |

Table 1: Closest-work comparison. EffectSlice is a downstream admission and audit layer, not a claim of superior compression or broader empirical coverage.

For registered block  $r$ , condition  $C \in \{B, F, S\}$  starts a fresh conversation with the same task, an isolated copy of the same starter workspace, action interface, public test, action budget, and private scorer. Only the artifact context differs. Let  $q_{C,r} \in [0, 1]$  be the terminal private score,  $t_{C,r}$  indicate the hard task contract, and  $p_{C,r}$  indicate that the terminal patch was applied. Operational success is

$$z_{C,r} = \mathbb{I}[t_{C,r} = 1 \wedge p_{C,r} = 1 \wedge q_{C,r} \geq 0.95]. \quad (2)$$

A threshold of 0.95 over 64 deterministic cases requires at least 61/64 cases. The raw scorer’s `success` flag means exact 64/64 success; every success count in our tables and decisions instead uses the operational definition above. Candidate rows use *Admit/Reject*; control and integrity checks use *Pass/Fail*. Provider or transport exhaustion, scorer failure, an empty submission, or a missing or malformed registered output makes the schedule *Invalid/No* record and supplies neither admission nor rejection evidence. When a valid scorer result exists but the terminal patch was not applied or the hard contract fails,  $z_{C,r} = 0$  and the run remains in the registered denominator.

EffectSlice makes a finite engineering decision. For an 18-block family, let  $k_C = \sum_r z_{C,r}$ . The candidate is admitted exactly when

$$\begin{aligned} k_B \leq 2, \quad k_F \geq 16, \quad k_S \geq 16, \\ k_F - k_S \leq 2, \end{aligned} \quad (3)$$

and every registered output and schedule-level integrity check is complete. Equation 3 is a Git-committed, hash-anchored service-level rule for this schedule. For a complete valid schedule that misses this gate, the record says *Reject*: it withholds admission under this frozen boundary rather than asserting that the candidate is semantically invalid. The binary decision applies only after the completeness and integrity gate; malformed schedules produce no decision record. It is not a confidence interval, prevalence estimate, or hypothesis about a population of papers or provider sessions.

The four numeric fields use a uniform two-run tolerance chosen from prior development evidence and frozen before V4; they are deployment parameters, not universal constants or values fitted to V4 outcomes. At  $n = 18$ , the registered

$k_F - k_S \leq 2$  field is logically implied by  $k_S \geq 16$  and  $k_F \leq 18$ . We retain it because it was preregistered, but do not count it as independent evidence. Admission is therefore a count-level SLA, not block-level behavioral equivalence.

The evidence-carrying output is a record

$$\mathcal{E} = (d_{\text{pre}}, M_{\text{post}}, \sigma, \mathcal{O}, \delta), \quad (4)$$

where the pre-run commitment root  $d_{\text{pre}}$  recursively binds  $d_P$ , the atom map and graph, rendered artifacts, task, public and private tests, scorer, bound analyzer, runner, transport, schedule  $\sigma$ , and their own digests. After execution, the digest-indexed archival manifest  $M_{\text{post}}$  binds  $d_{\text{pre}}$ , immutable output set  $\mathcal{O}$ , raw-evidence digest, analysis and compatibility code, independent crosscheck, generated result macros, and local decision  $\delta$ . The supplied `effectsliceslice-paper-results-manifest.v5` file `generated_results_manifest_v5.json` is canonical; older numbered manifests are superseded. It preserves lineage but does not retroactively preregister post-run repairs or derived artifacts. A score without these candidate, commitment, and information-boundary identities is not an EffectSlice admission record.

## EffectSlice

### Source-Addressed Candidate Construction

The extraction stage creates the atom registry from quoted paper passages and records dependencies separately from prose order. A reducer may be heuristic, learned, or interactive; the admission protocol only requires a deterministic candidate identity and closure audit. SNAP prefix 03 and Toolformer prefix 01 were locked before V4 provider calls. Toolformer prefix 04 already existed in the same deterministic reducer registry, but we selected it after inspecting V4, then froze it before V5. This project-local ordered prefix reducer is neither our contribution nor an implementation of SkillReducer.

The closure auditor rejects unknown atom IDs, graph cycles, a non-strict set, or any retained atom whose transitive prerequisite is absent. It then renders with the registered topological tie break and verifies  $d_F$  and  $d_S$ . Every V4 and V5 strict-subset family passed these checks before its

provider execution. The atom boundaries and dependency edges were author-engineered from the cited passages; their semantic correctness was not independently adjudicated.

Table 2 exposes the complete atom-ID inventory. The SNAP candidate retains preprocessing and degree construction but omits the linear-operator and eigensolver/guard atoms. Toolformer prefix 01 retains only the future-token weight schedule; prefix 04 retains the loss, counterfactual, and threshold but omits the all-candidate/order unit. They are concrete core hypotheses. Prefix 04 was selected using V4 results, so its claim rests only on the subsequently registered V5 block, not on prospective V4 selection.

## Public Development, Terminal Private Scoring

The model sees the task, common action protocol, assigned artifact, an isolated starter-workspace copy, and the public test file. Within a 16-action budget it may search, open, edit, and run the public test. Public output is returned and may guide another model turn. A `submit` action freezes the patch, invokes the private 64-case scorer once, stores the result, and terminates the conversation. If the action budget ends with a nonempty patch, that final patch is scored once; an empty patch fails without feedback. No private result is sent back to the model.

The family manifest binds the paper text, atom map,  $F$ ,  $S$ , public test, private case registry, scorer, starter workspace, prompt, runner, analyzer, transport, thresholds, block identifiers, and B/F/S order by SHA-256. Output directories are write-once. Provider response IDs must be unique, and every transport attempt is retained. This yields a trace from a paper span to the artifact seen by the model, its tool trajectory, the fixed patch, and the final score.

## Controls and Interleaved Schedule

V4 uses 18 temporal strata and at most two workers. Every stratum contains SNAP, Toolformer prefix 01 (also the negative control), and the planted positive; six also contain identity instrumentation. Each 18-block family balances all six B/F/S orders three times, while identity uses each once: 60 blocks and 180 fresh conversations in total.

After V4, we selected pre-existing prefix 04. V5 separately freezes 18 balanced blocks and 54 fresh conversations over cases disjoint from all earlier private cases by ID, seed, and payload digest. It is new evidence for a post-V4 candidate, not an extension of the V4 preregistration.

The planted family is a pipeline smoke test, not evidence of useful compression or naturally redundant paper text. The registered internal checks pass only if it is admitted, the real negative is rejected, identity instrumentation passes, all outputs exist, and integrity holds. Identity passes when at least five of six F/S success indicators match and the aggregate F/S success-count gap is at most one. At six pairs the latter is implied by the former and is retained as a registered consistency field, not independent evidence. Identity does not require the two fresh sessions to return the same patch or score.

## Experimental Setup

**Paper-specific tasks.** SNAP-MFSE isolates SnapATAC2’s matrix-free spectral embedding core (Zhang et al. 2024): IDF scaling, row normalization, self-similarity removal, degree construction, normalized operator application, and leading spectral coordinates. Toolformer filtering isolates the rule for retaining API calls (Schick et al. 2023): future-token weights, weighted losses, the stronger counterfactual, inclusive thresholding, and ordered application to all proposals. Each V4 or V5 private block contains 64 deterministic cases and hard interface/guard checks. Cases characterize one terminal patch and are not treated as independent model runs.

**Freeze and holdout chronology.** V3 development evidence informed the candidates and the uniform two-run SLA tolerance. We then froze  $F$ ,  $A_S$ ,  $S$ , both controls, thresholds, and the 18-stratum schedule in Git commit `ae97dfff`, before any V4 provider call. The V4 builder generated new private case blocks and rejected overlap with every earlier private block by case ID, seed, and case-payload digest. V4 private payloads did not select its candidates. Visible public tests were fixed before private execution but are development feedback, not holdout evidence.

After observing V4 outcomes, we selected Toolformer prefix 04 from the pre-existing reducer registry and preregistered V5 in commit `da0ee5d`. Attempt 1 stopped during local preflight because the generic runner did not yet allow the new case-block label; it made zero provider calls and wrote no block outputs. Commit `6538e8d` added that explicit allowlist entry, froze a new evidence boundary and preregistration digest, and preceded every V5 provider call. The V5 builder again excludes overlap by ID, seed, and payload digest.

**Provider and host boundary.** Inference used the user-supplied trusted DeepSeek endpoint. We report the provider/user label “DeepSeek V3.2” and exact request alias `deepseek-v4-flash`; the alias-to-deployment mapping is not independently attested. Requests used OpenAI-compatible chat completions, temperature zero, an 8192-token turn limit, a 240-second timeout, and at most five transport attempts. The provider exposes no supported random seed, so “fresh” means a new top-level conversation, not independent or reproducible backend sampling.

A Windows PC with a Ryzen 9 7945HX, 31.7 GiB RAM, and an 8 GiB RTX 4060 Laptop GPU performed orchestration, bounded execution, testing, scoring, and persistence. Model inference stayed at the vendor API; the local GPU was not used. This is an API-first ordinary-device setting, not a local-quantized-model result.

## Results

### Three Evidence Questions

Figure 1 and Table 3 report both complete registered schedules.

**RQ1: Do B/F/S runs separate two kinds of insufficiency?** In V4, SNAP B/F/S succeeds in 0/15/5 blocks. The full

| Task/family          | Registered source-addressed units in $F$                                                                                                                                                                                                                                                                              | Fixed $S$     |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| SNAP-MFSE            | A01 IDF scaling; A02 row normalization; A03 self-similarity removal and degree; A04 matrix-free linear operator; A05 eigensolver, ordering, and no-materialization guard. <i>Source/deps: extracted lines 73–79; A01 <math>\prec</math> A02 <math>\prec</math> A03 <math>\prec</math> A04 <math>\prec</math> A05.</i> | A01–A03 (3/5) |
| Toolformer prefix 01 | T01 normalized future-token weights; T02 weighted loss; T03 stronger counterfactual; T04 inclusive threshold; T05 all-candidate/order rule. <i>Source/deps: lines 76–79, 134–171, 268–272; T01 <math>\prec</math> T02 <math>\prec</math> T03 <math>\prec</math> T04 <math>\prec</math> T05.</i>                       | T01 (1/5)     |
| Toolformer prefix 04 | T01–T05 and source/dependency chain as above. This candidate was a pre-existing output of the registered deterministic prefix reducer.                                                                                                                                                                                | T01–T04 (4/5) |
| Planted positive     | T01–T05 above plus T06, a registered restatement of the T04/T05 invariant that adds no computation. <i>T06 cites lines 76–79 and 159–171 and requires T04/T05.</i>                                                                                                                                                    | T01–T05 (5/6) |

Table 2: Complete atom-ID inventory and explicit candidate boundary. The bound registry retains each atom’s full locator, instruction, role, and dependency fields, plus byte spans and file digests. The identity family uses byte-identical  $F$  and  $S$  and instrumentation, not a strict-subset candidate.

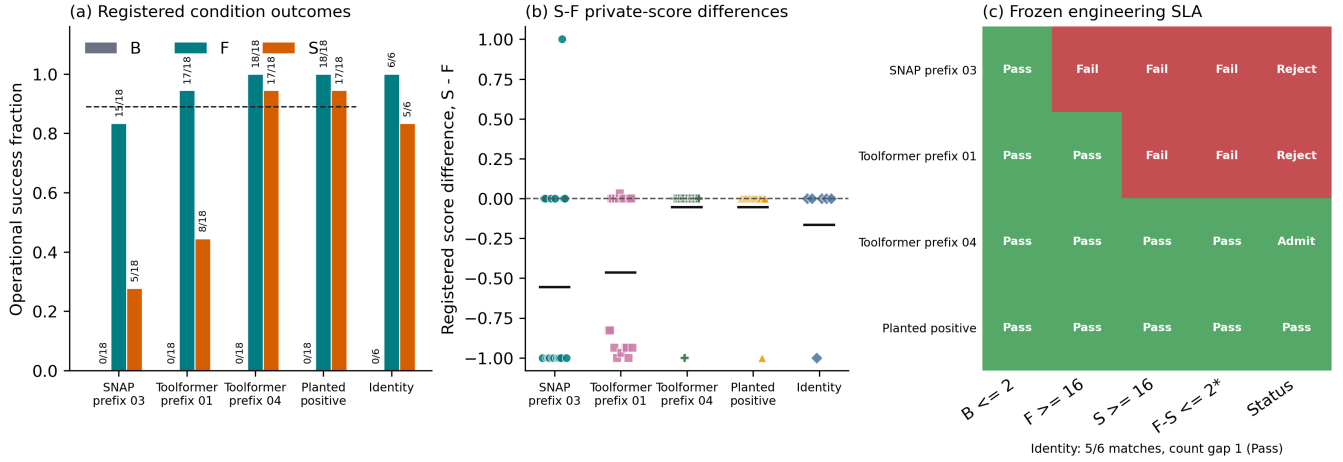


Figure 1: Complete V4 evidence plus the separately registered, later V5 candidate. (a) B/F/S operational successes; labels show exact counts, and the dashed line marks the 16/18 requirement for the four 18-block strict-subset families. (b) Per-block  $S - F$  private-score differences; black lines are family means, not population estimates. (c) Frozen SLA fields and decisions. The starred field is implied by  $S \geq 16$  at  $n = 18$ ; identity uses a separate rule.

artifact misses the minimum by one, the slice misses it by eleven, and the  $F-S$  count gap is ten. The result therefore finds that neither  $F$  nor  $S$  meets its frozen sufficiency requirement for this schedule.

For Toolformer prefix 01, B/F/S succeeds in 0/17/8 blocks. Here the full artifact clears its requirement, but the one-atom slice does not and trails  $F$  by nine successes. The two realized patterns therefore distinguish full-artifact insufficiency (SNAP) from subset-only insufficiency (Toolformer prefix 01) under their registered schedules.

**RQ2: Do the controls exercise the intended sanity states?** The planted-positive control is admitted, this registered negative control is rejected, and byte-identical identity instrumentation passes despite one fresh-session mismatch. Thus all three intended internal sanity states are exercised; they are not estimates of population-level operating characteristics.

**RQ3: Can a post-selected candidate earn a valid local admission on a separately registered, nonoverlapping**

**boundary?** Prefix 04 existed before V4 but was selected after V4 analysis; V5 then froze and evaluated it on new private cases. It retains T01–T04 and omits T05. B/F/S succeeds in 0/18/ 17 blocks, with means 0.000/1.000/ 0.944. All six frozen SLA and integrity fields pass, so the non-planted strict subset is admitted for this task, provider, and private registry. The result does not imply that T05 is dispensable for Toolformer in general; it says this fixed artifact substitutes under the registered V5 boundary.

The omitted atoms offer a plausible diagnosis but not a causal ablation. SNAP’s slice stops before the matrix-free operator and solver guard; Toolformer prefix 01 stops before every loss and decision unit. V4 failures are consistent with those omissions, but no atom-restoration ablation identifies a cause. Prefix 04 omits only the all-candidate/order unit and passes V5; that is task-local sufficiency evidence, not a causal or paper-wide redundancy claim.

| Family                   | Blocks | B success | F success (mean $q$ ) | S success (mean $q$ ) | F-S gap | Decision/status |
|--------------------------|--------|-----------|-----------------------|-----------------------|---------|-----------------|
| SNAP real                | 18     | 0         | 15 (0.877)            | 5 (0.321)             | 10      | Reject          |
| Toolformer real/negative | 18     | 0         | 17 (0.944)            | 8 (0.481)             | 9       | Reject          |
| Toolformer prefix 04     | 18     | 0         | 18 (1.000)            | 17 (0.944)            | 1       | Admit           |
| Planted positive         | 18     | 0         | 18 (1.000)            | 17 (0.944)            | 1       | Pass            |
| Identity instrument      | 6      | 0         | 6 (1.000)             | 5 (0.833)             | 1       | Pass            |

Table 3: Registered outcomes across sequential V4 and V5 schedules. Admission uses the frozen count-level SLA plus complete output and integrity. The F-S field is retained but redundant at these thresholds. Identity is not an admission and uses its separate 5/6 match rule.

## Registered Internal Sanity and Integrity Checks

The preregistered `expected_admission` fields are expectation metadata only: they encode internal-check expectations for controls and planning expectations for other candidates. They never enter the admission decision, which is recomputed solely from registered outputs and SLA fields.

The planted family removes only redundant T06. It yields B/F/S counts 0/18/17, an F-S gap of one, and passes every registered field. The same scorer and Toolformer task reject the real one-atom candidate at 0/17/8. Thus the protocol does not reject every strict subset, and its registered positive and negative expectations are both observed. This is a harness smoke test, not evidence that a non-planted compact core has been found. That stronger, still task-local evidence comes from V5 prefix 04: it was present in the reducer registry before V4, selected after V4, frozen before V5, and admitted on new private cases. This chronology avoids V5 holdout reuse while openly retaining post-V4 selection bias.

For identity instrumentation, F and S are byte-identical yet their fresh conversations succeed in 6/6 and 5/6 blocks. Five of six paired success indicators match and the aggregate gap is one, so the instrument passes. The single mismatch is important: artifact identity does not imply trajectory or patch identity under fresh provider sessions. This motivates a tolerance-based finite SLA rather than exact per-pair equality.

Together, planted-positive admission, real-negative rejection, and identity instrumentation make the registered internal-check status Pass (the macro retains the historical field name). This is a task-local sanity and integrity check of the measurement stack, not sensitivity or specificity over a population of skills.

## Public Iteration and Evidence Cost

The two schedules contain 234 conversations, 1760 provider turns, and 1760 recorded actions. The agent requested public tests 291 times; 278 passed and 13 did not. V5 alone contains 381 actions, 17 invalid actions, one nonpassing public test, and 2 extra transport attempts. Every V5 conversation explicitly submitted; all V4 and V5 outputs are complete.

The provider reported 2,812,362 input and 2,894,613 output tokens across 1,776 transport attempts. These totals document evidence cost, not an efficiency advantage. The protocol deliberately spends repeated API calls to audit substitution rather than claiming a shorter skill is cheaper.

## Analyzer and Release Integrity

The V4 and V5 bound analyzers stopped before aggregation because a copied check conflated `comparison_role=development_triage` with the formal `evidence_boundary`. Narrow post-hoc adapters validate both raw fields, normalize only that in-memory comparison, and leave preregistrations, runners, transcripts, patches, scores, thresholds, schedules, and raw manifests unchanged. Independent outcome/count crosschecks import neither analyzer nor adapter. They reread 660 V4 and 198 V5 schema-selected canonical raw-evidence files, verify terminal private-score and response-ID invariants, and reproduce every reported count, mean, resource total, control, and decision. The two archived result-root snapshots contain 1096 files in total. The canonical selection comprises block-level `pair_manifest.json` and `run_report.json`, plus condition-level `candidate.patch`, `run_result.json`, and `transcript.json`; auxiliary duplicate scorer patches and progress/lock files are excluded. The canonical post-run manifest records its pre-run roots, both raw inventories, analysis lineage, tests, and generated results. A separate release verifier enumerates exactly those 858 schema-selected canonical raw-evidence files, requires set equality with the inventories, and rehashes them plus 24 directly bound artifacts. These checks do not rerun the scorer or independently validate the semantic task contract.

## Discussion

**Source-addressed provenance is needed but is not sufficiency.** A source span can prove that an instruction came from the paper. It cannot prove that a selected set contains every operation an agent needs or that the agent will execute them reliably. EffectSlice connects provenance to a behavioral record without conflating them: source atoms answer “why is this instruction here?”, while the B/F/S schedule answers “may this fixed subset substitute for the complete artifact here?”

**The contribution is admission, not compression.** SkillReducer’s self-correcting optimizer can restore content after task failures (Gao et al. 2026); Xu and Wu show that representation changes can be dominated by executor capability (Xu and Wu 2026). Either pipeline could feed a candidate into EffectSlice. Our result does not overturn their aggregate findings. It shows that one compact candidate

should carry a source identity, an information-flow boundary, and a local admit/reject record before reuse.

### **A local admission or rejection is the reusable output.**

For a nonexpert workflow, the useful output need not always be a smaller skill. If F is not sufficiently reliable, retain the full artifact with a warning. If F works but S fails, expose which source units were omitted and do not advertise S as the paper’s core. If S passes, publish it together with its task, provider, scorer, schedule, and evidence boundary. This defines an auditable output contract; whether it helps people complete real projects faster remains an open user-study question.

## **Limitations and Broader Impact**

The study covers two papers, two bounded implementation tasks, one provider endpoint, one action scaffold, and two sequential registered schedules. The tasks exercise paper-specific mechanisms rather than complete SnapATAC2 analyses or Toolformer training. The simple prefix candidates are intentionally weak reducers; we neither run nor claim an official external reducer implementation. The positive control removes planted redundancy. Broader candidate generators, papers, tasks, providers, and independently designed registries are required before any effectiveness claim. The one admission among three non-planted candidates is only a descriptive count across sequential schedules, not a prospectively registered rate or aggregate performance result.

The atom boundaries, executable instructions, dependency graph, and the claimed completeness of F were not independently reviewed. Their digests prove identity, not semantic correctness. The two V4 candidates have structurally obvious omissions. V5 admits one closer, non-planted Toolformer subset, but it was selected after V4 and tested on only one new registry for the same task. This is not cross-paper or paper-wide equivalence. Finally, the count-level SLA does not establish block-level trajectory preservation.

Fresh conversations are not verified independent samples. The provider exposes neither a seed nor backend routing guarantees; caching, updates, and shared infrastructure may induce dependence. V4 and V5 therefore report only their complete finite schedules. Their shared thresholds are fixed engineering choices, not estimated operating characteristics. Each decision remains conditioned on its particular private registry.

The visible public test is intentionally useful and therefore can be overfit. The terminal private suite limits direct score adaptation but cannot guarantee semantic generalization beyond its 64 cases. Independent task-contract review, additional hidden registries, and adversarial cases would strengthen the record. The analyzer repairs are fully disclosed and outcome/count cross-checked, but remain post-registration changes to the analysis path. V5 repeated the same role-field defect already found in V4, exposing a regression-testing failure even though the narrow adapters and crosschecks preserve the reported decisions. The release supports bit-level audit and offline count recomputation. It

does not guarantee an end-to-end provider-trajectory rerun: raw pair manifests retain historical absolute paths, and the provider seed, alias-to-backend mapping, and backend state are not independently reproducible. Crosschecks do not rerun the scorers or independently validate the semantic task contract.

No human participants were studied. We do not measure understanding, time to reproduce a paper, or downstream scientific correctness. Evidence-carrying admission can reduce overconfident reuse, but a narrow pass can also create false assurance if its paper, task, provider, and scorer boundaries are omitted. Generated code remains a supply-chain risk; bounded workspaces and private scoring reduce but do not eliminate it.

## **Conclusion**

EffectSlice treats a paper-derived core as a candidate substitution claim. It binds source-addressed provenance, visible public development, terminal private scoring, and registered internal checks into a local decision. Across 180 V4 conversations, controls pass, SNAP’s registered full artifact misses sufficiency, and Toolformer prefix 01’s slice misses despite sufficient F. A later schedule admits non-planted prefix 04 at 0/18/17 B/F/S successes on fresh registered evidence. The contribution is a proof-of-concept protocol and evidence package for an auditable local admit/reject decision, not universal compression.

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